

DEEPAK KUMAR SINGH

Embedded Systems & Platform Software Engineer · C/C++ · Embedded Linux · Python
+91 7880414295 | deepakausingh@gmail.com | linkedin.com/in/deepakdan | Varanasi, Uttar Pradesh, India

PROFESSIONAL SUMMARY

Computer Science graduate with strong fundamentals in C/C++, embedded systems, and low-level software development. Hands-on experience with hardware-interfacing, real-time pipelines, and ML-driven embedded applications. Proficient in Python and shell scripting; familiar with embedded Linux. Keen interest in AI-assisted engineering tools for code quality, debugging, and productivity improvement.

TECHNICAL SKILLS

Languages	C, C++, Python, Shell Scripting, JavaScript
Embedded	Embedded Linux, Memory Management, Interrupts, Concurrency, GPIO, UART, SPI, I2C, PCIe basics
Platform/BSP	Board Bring-Up, Device Drivers, Boot Initialization, Platform Services, Subsystem Integration
Networking	TCP/IP Fundamentals, Ethernet, RESTful APIs, Wireless Protocols (basic)
ML / Tools	TensorFlow, OpenCV, MediaPipe, Scikit-learn Git, GDB, VS Code, CI/CD

WORK EXPERIENCE

Software Developer Intern — Extion Infotech, Agra, India

June – Aug 2024

- Built ML-based data processing pipeline in Python & TensorFlow — applying low-level data ingestion and validation techniques analogous to embedded sensor pipelines.
- Collaborated across hardware-adjacent and software teams in agile workflows; upheld production-grade code review and CI standards.

Web Development Intern — Udemy (Remote)

Sep 2024

- Completed full-stack MERN training; practised system integration, API design, debugging, and automated build pipelines — skills transferable to embedded platform development.

KEY PROJECTS

Embedded Linux GPIO Controller Simulation C · Python · Embedded Linux · Shell Scripting

- Implemented GPIO pin control in C on embedded Linux: interrupt-driven I/O, memory-mapped registers, concurrent threads (pthreads); debugged via GDB across I2C and SPI interfaces.
- Wrote shell scripts to automate build-flash-test cycles, simulating a BSP bring-up workflow for a multi-peripheral board.

Real-Time Human Activity Detection Python · TensorFlow · OpenCV · Scikit-learn

- Built a real-time ML inference pipeline for activity classification from video/sensor streams; optimized memory footprint and latency to approach embedded deployment constraints.
- Integrated automated regression testing scripts, improving validation coverage and debug turnaround.

3D Gesture Interface — Embedded HID Simulation Python · MediaPipe · OpenGL · PyOpenGL

- Developed a hardware-interfacing system translating webcam gesture inputs into 3D commands via low-latency acquisition loops with noise-reduction — emulating interrupt-driven embedded HID firmware.

Platform Service Monitor Daemon C · Python · Shell · Embedded Linux

- Built a C daemon monitoring CPU/memory/process states on embedded Linux with GPIO-triggered alerts, watchdog resets, and safe shutdown sequences — mirroring platform service reliability requirements.

EDUCATION

B.E. in Computer Science — Dr. Bhimrao Ambedkar University, Agra 2021 – 2025 | CGPA: 7.5

Coursework: Data Structures & Algorithms, OOP, C/C++, Python, AI & ML, Computer Architecture, Operating Systems, DBMS

Intermediate (10+2) — Shandilya Public School, Varanasi

2018 – 2020

Distinction in Mathematics and Science | Percentage: 72%

High School (10th) — Shandilya Public School, Varanasi

2017 – 2018

Distinction in Mathematics and Science | Percentage: 84%